

Resilient decentralized communication for unstructured peer-to-peer networks

From Sparsity to Reliability: the Decentralized Path to Free Speech

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Introduction

Motivation

Provide resilient distributed communication during Internet shutdowns in case of disasters

- Consolidation of power in Internet infrastructure due to running services, protocol, and algorithms in a few closed providers
- Centralization enables both rapid information diffusion but also censorship, failure points, and control, worsened by disasters and political unrest



Towards addressing fragile consolidation of power, resilient decentralization is key!

- Providing connectivity without infrastructure: p2p networks
 - Structured: Kademlia [MM02], Chord [SMK⁺01], more recently DLT [GTO24]: ETH [GTDO24], Bitcoin [Nak08], Ripple XRP [GTMF22], etc
 - **Unstructured**: FireChat [Sim14], Twimight [Leg11], Briar [bri18], bitchat [JD25], Amigo [IRJ⁺25]

To enable sustainable critical communication for all: use efficient ubiquitous p2p protocols

- **BLE**
- Others: WiFi-Direct, Zigbee, LoRA [FKK13]

Introduction

Contribution

Research Question How to provide resilient decentralized communication over unstructured p2p networks using efficient p2p communication protocols in the presence of network disruptions?

Approach Collect empirical data to parameterize signal, simulate distributed network topologies, measure resilience w.r.t centralization metrics

Introduction

State-of-the-Art

Setup

Evaluation

Discussion

Future Work

- Distributed resilience in communication, not resources: overall similar node degrees in topology
- Compare different metrics: L_0 norm, $\mathcal{L}$ -Score [HRAD25], Gini [Nic08]

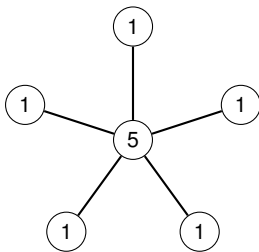


Figure 1: Star topology, $L_0 = 1.0$

State-of-the-Art

Resilience Metrics: the $\mathcal{S}$ -Score

- Metric coming from US antitrust laws to measure market concentration
- Measures similarity to maximally centralized topology

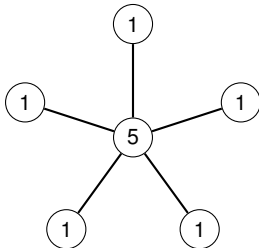


Figure 2: Star topology, $\mathcal{S} = 1.0$

State-of-the-Art

Resilience Metrics: Six Criteria of Sparsity [Nic08]

1. Robin Hood: Stealing from the rich and giving to the poor decreases sparsity
2. **Scaling**: multiplying all values by a constant does not change sparsity
3. Rising Tide: Adding a constant to all values decreases sparsity
4. Cloning: Cloning the vector does not change sparsity
5. Bill Gates: Adding a large value to the vector increases sparsity
6. Babies: Adding a zero to the vector increases sparsity

- $\mathcal{S}$ -Score does not satisfy **Scaling** axiom

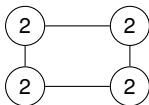


Figure 3: Topology $[2, 2, 2, 2]$ with $\mathcal{S} = 0.75$

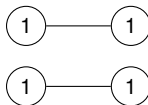


Figure 4: Topology $[1, 1, 1, 1]$ with $\mathcal{S} = 0.5$

State-of-the-Art

Resilience Metrics: the Gini Coefficient

- Popular metric used to compare wealth distributions of populations
- Measures inequality of node degrees in a topology

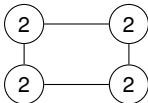


Figure 5: Topology [2, 2, 2, 2] with $G = 0$

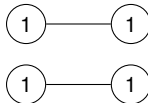


Figure 6: Topology [1, 1, 1, 1] with $G = 0$

Introduction

State-of-the-Art

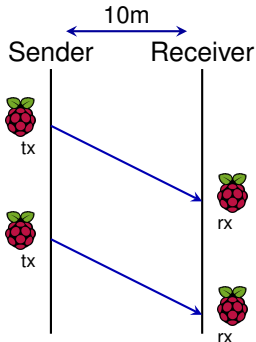
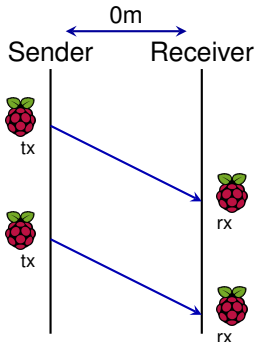
Setup

Evaluation

Discussion

Future Work

- **What** Measure real-life outdoor signal propagation of BLE: measure RSSI of BLE advertisements at increasing distances until failure
- **How** Two standard Raspberry Pi 3B+ boards, open field with minimal interference
- **Why** Obtain empirical data to parameterize simulations of BLE communication at scale



- Rx pi scan window 600s, tx pi advertisement interval 20ms, repeat 20 times
- Measured RSSI at distances: 0m, 1m, 5m, 10m, incrementing by 10m until no more ads received: 100m

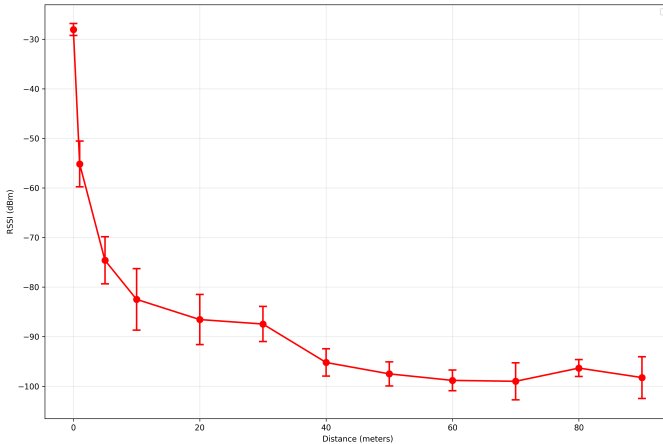


Figure 7: Mean RSSI vs. Distance

- Model advertisement success probability vs. distance using empirical RSSI data
- Use number of advertisements received at $d_0 = 0$
- Probability of advertisement being received at distance d :

$$P_{adv}(d) = \frac{\text{Number of received advertisements at } d}{\text{Total number of received advertisements at } d_0}$$

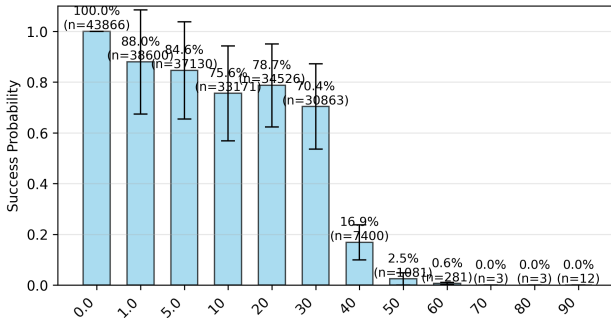


Figure 8: Advertisement Success Probability vs. Distance

Setup

Simulations

- **What** Measure resilience of communication clusters established using empirical BLE signal stochastics
- **How** The ONE Simulator, $400m \times 400m$ L-shaped field, divided to $100m \times 100m$ grid cells

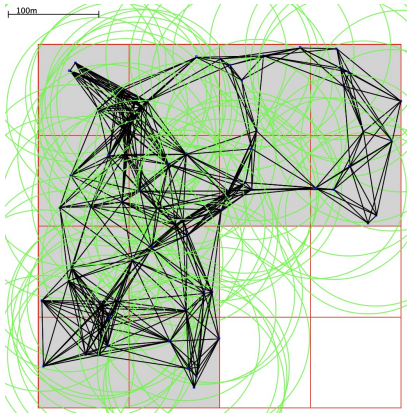


Figure 9: Simulation Area

- 12 cluster cells, 6 hosts per cell, static movement model
- Epidemic Router, same amount of messages sent between each host pair
- 25 simulator runs consist of: network interface communication range, maximum peer degree, communication mode

- Parameterize network interface bandwidth based on link distance using RSSI results from testbed
- Use log-normal path loss model [ASC23] to interpolate signal propagation decay between discrete distance points
- BLE 1M PHY [She21]: Use maximum channel bandwidth of 1 Mbps at $\leq 1m$ distance to deduce message transfer time

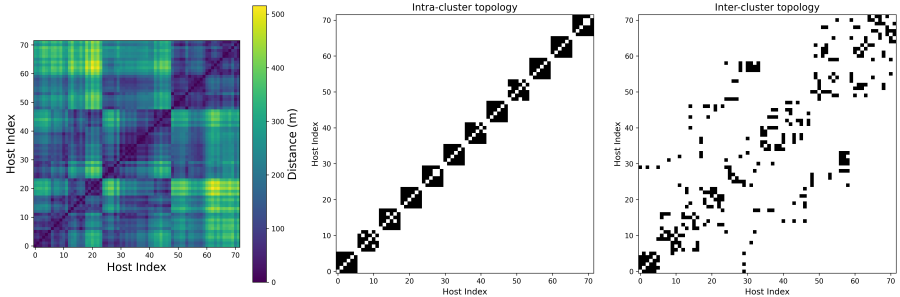


Figure 10: Topological Matrix for communication range 90m and max peer degree 5

Introduction

State-of-the-Art

Setup

Evaluation

Discussion

Future Work

Evaluation

Model Insights

- Gini much more indicative of resilience, as $\mathcal{S}$ – Score saturates very quickly, not indicative of sparsity
- Centralization correlates with an increase in peer degree, especially in higher ranges
- In lower communication ranges, increasing the maximum node degree has almost no effect on centralization

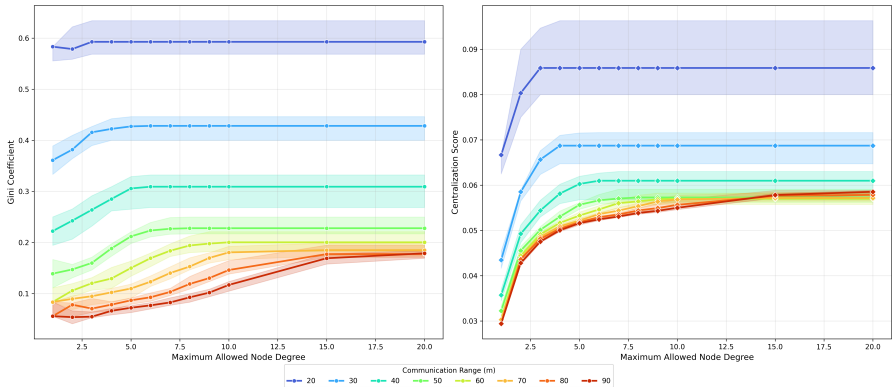


Figure 11: Centralization vs. Max Peer Degree

Model Insights

- Increasing the maximum peer degree beyond a threshold decreases latency for long-distance messages
- Short-distance messages are penalized by a high peer degree due to increased contention
- Long tail of short-distance indicates most direct links are short ranged

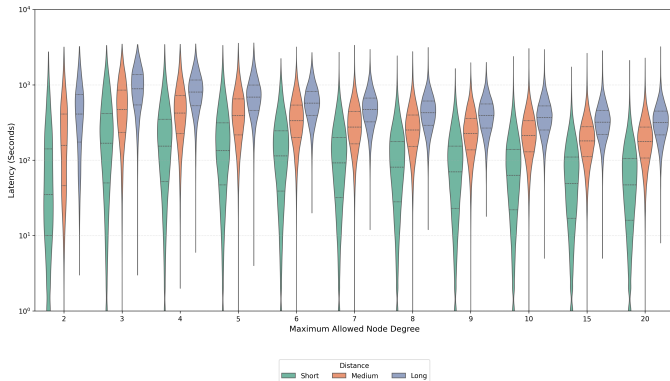


Figure 12: Latency vs. Max Peer Degree at BLE 1M PHY Range

- Latency decreases with increasing centralization for long-distance messages
- Increasing centralization for short-range messages penalizes latency due to contention

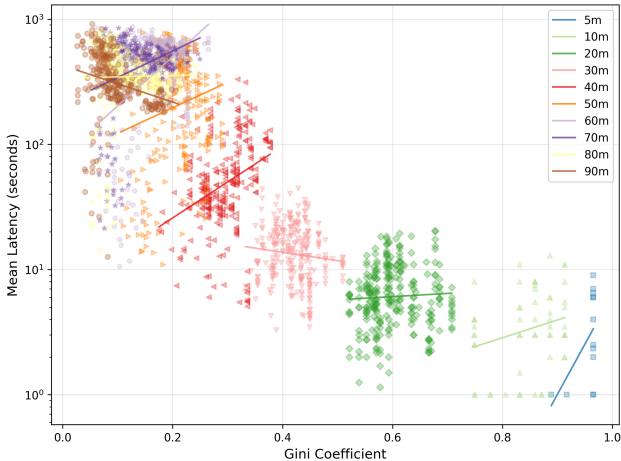


Figure 13: Latency vs. Gini Coefficient

- Latency decreases with increasing centralization for long-distance messages
- Increasing centralization for short-range messages penalizes latency due to contention

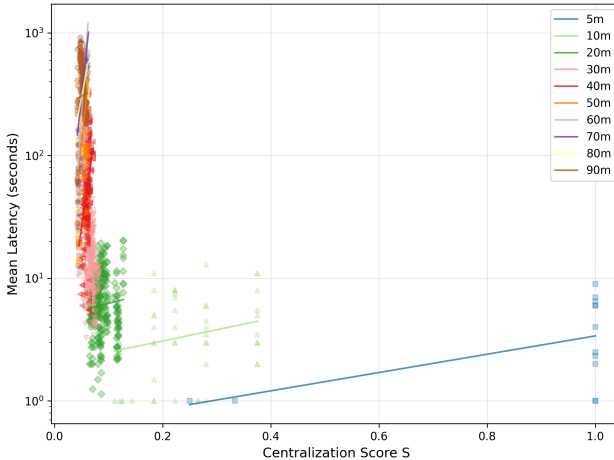


Figure 14: Latency vs. $\mathcal{S}$ — Score

Introduction

State-of-the-Art

Setup

Evaluation

Discussion

Future Work

- Gini coefficient is a better indicator of resilience than the $\mathcal{I}$ -Score
- $\mathcal{I}$ -Score much more sensitive to extreme centralization, detecting SPOFs
- Design considerations for establishing P2P BLE networks: consider centralization and intended message range when designing P2P networks, communication-intense algorithms reduce resilience
- Gossip without consideration of peer degree can lead to communication bottlenecks and reduced resilience: consider global goal parameters such as resilience when designing p2p communication algorithms!

Introduction

State-of-the-Art

Setup

Evaluation

Discussion

Future Work

- Explore other resilience metrics, e.g. Freedman Centralization Score
- Introduce edge churn based on advertisement success probability plot
- Measure throughput and latency in testbed based on BLE transport (unicast P2P instead of broadcast)
- Repeat empirical experiment for longer time, use the same environment for low and high ranges, compare different times of day, use custom Bluetooth stack for more control (control channel)
- Repeat experiment using mobile phones
- MA: Build p2p peering algorithm based on design considerations from this work, measure resilience in practice

Thank you!
Questions?



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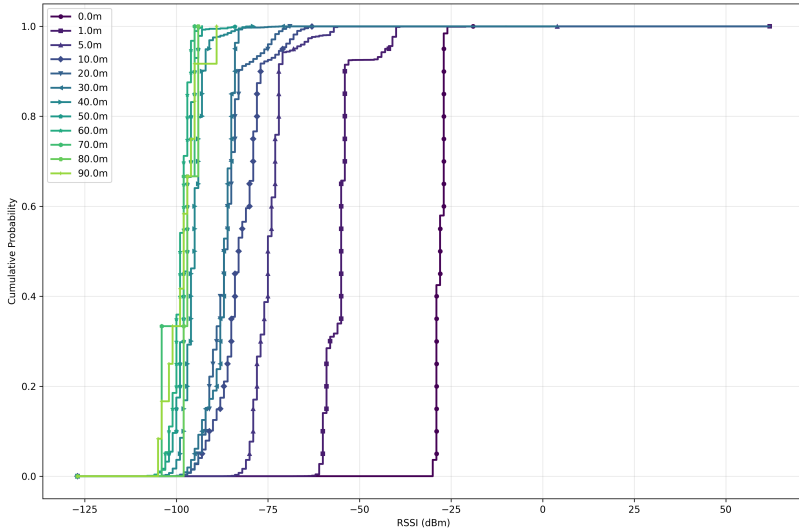


Figure 15: RSSI CDF